

UNIT - 1

TAYLOR'S SERIES

A Taylor's Series is a series expansion of a function about a point.

$$f(x+h) = f(x) + hf'(x) + \frac{h^2}{2!} f''(x) + \frac{h^3}{3!} f'''(x) + \dots$$

$\Downarrow$ (special case) $\left[\begin{array}{l} x=0 \\ h=x \end{array} \right]$

$$f(x) = f(0) + xf'(0) + \frac{x^2}{2!} f''(0) + \frac{x^3}{3!} f'''(0) + \dots$$

for instance : Taylor expansion of $f(x) = e^x$.

$$e^x = e^0 + xe^0 + \frac{x^2}{2!} e^0 + \frac{x^3}{3!} e^0 + \dots$$

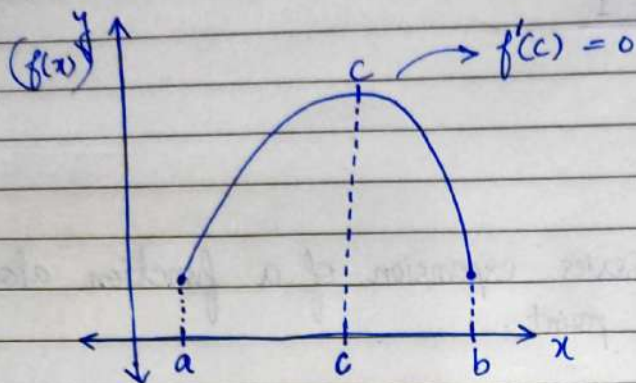
$$\Rightarrow e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

ROLLE'S THEOREM

- 1) If $f(x)$ is continuous on $[a, b]$.
- 2) If $f(x)$ is differentiable on (a, b) .
- 3) $f(a) = f(b)$ holds,

then, $\exists$ at least one $c \in (a, b)$ such that $f'(c) = 0$.

Graphical Representation of Rolle's Theorem.



Ques. $f(x) = 3 + (x-2)^{2/3}$ on $[1, 3]$ existence of Rolle's Theorem.

Ans. $f'(x) = \frac{2}{3} \frac{1}{(x-2)^{1/3}}$

so,

$$x-2 = 0$$

$$x = 2$$

so, derivative doesn't exist on $x=2$ and $2 \in (1, 3)$.
 $\therefore$ Rolle's Theorem doesn't hold.

Note : Polynomial, exponential, logarithmic, trigonometric, Inverse Trigonometric functions are always continuous & differentiable on their domain.

* Rolle's Theorem for POLYNOMIAL

- 1) If $f(x)$ continuous on $[a, b]$.
- 2) If $f(x)$ is differentiable on (a, b) .
- 3) $f(a) = f(b) = 0$ i.e. a & b are roots..

then, $\exists$ at least one root of $f'(c)$ with 2 roots of $f(x)$ ($a \neq b$) included.

MEAN - VALUE THEOREM

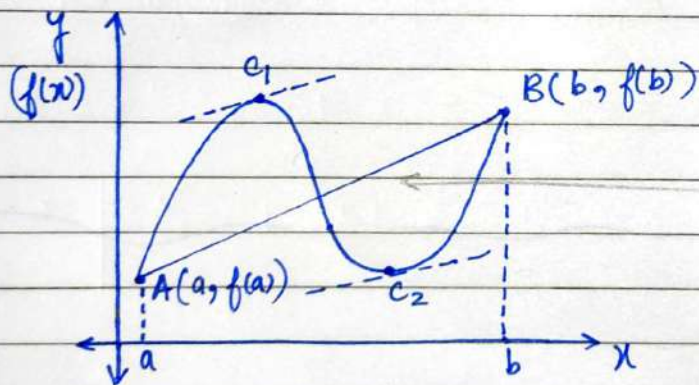
let $f(x)$ be a function which is :

- (1) continuous on $[a, b]$.
- (2) differentiable on (a, b) .

then, $\exists$ at least one point $c \in (a, b)$ such that :-

$$f'(c) = \frac{f(b) - f(a)}{b - a}$$

Graphical Representation.



Secant made by end points (A & B) of the curve

→ Tangents made on points c_1 & c_2 are parallel to the secant.

SIGNIFICANT FIGURES

Rules

- Non-zero digits are significant.
- Zeros b/w non-zero digits are significant.
- Final zeroes in the decimal portion are significant.
- Final zeroes are non-significant if there's no decimal point.

Ques.) find no. of Significant figures :-

1) 58	-	2
2) 603	-	3
3) 1008	-	4
4) 95080	-	4
5) 70	-	1
6) 0.80	-	2
7) 18.00	-	4
8) 0.0003	-	1
9) 0.0030	-	2

ROUNDING OFF

1) 39.72431	(3D)	-	39.724
2) 79.87893	(3D)	-	79.879
3) 2.589951	(2D)	-	2.59
4) 59.4865	(3D)	-	59.486
5) 47.6975	(3D)	-	47.698

UNIT - 1BISECTION METHOD

→ This method is based on the repeated application of the intermediate value theorem to obtain an approximate root of the equation $f(x) = 0$.

Choose 'a' & 'b' such that : $f(a) \rightarrow +ve$ $f(b) \rightarrow -ve$

↓ i.e.

$$f(a) \cdot f(b) < 0$$

then, a root lies b/w a and b. The mean value is taken as the first approximate value of the root.

$$x_0 = \frac{a+b}{2}$$

- If, $f(x_0) = 0$, then obviously $x = x_0$ is a root of the equation $f(x) = 0$.
- But if, $f(a) \cdot f(x_0) < 0$ i.e. $f(a) \rightarrow +ve$ & $f(x_0) \rightarrow -ve$

then, the next approx. root lies in the interval (a, x_0) .

OR if, $f(b) \cdot f(x_0) < 0$ i.e. $f(b) \rightarrow -ve$ & $f(x_0) \rightarrow +ve$.
then, the next approx. root lies in the interval (x_0, b) .

Note : The new interval is of the length $\left(\frac{b-a}{2}\right)$.

Now,

- If the root lie in the interval (a, x_0) , then the new approximate root will be :-

$$x_1 = \frac{a + x_0}{2}$$

- If the root lie in the interval (x_0, b) , then the new approx. root will be :-

$$x_1 = \frac{x_0 + b}{2}$$

- If $f(x_1) \cdot f(a) < 0$ i.e. $f(a) \rightarrow +ve$ & $f(x_1) \rightarrow -ve$, then the next approx. root (x_2) will lie in the interval (a, x_1) .

Otherwise, the interval will be (x_1, x_0) .

Note : The length of this interval will be $\left(\frac{b-a}{2^2}\right)$.

→ This process is repeated until the interval (which contains the root) is as small as desired, say numerically less than some given small number $\epsilon > 0$.

In General :-

The length of the interval at the n^{th} step will be $\frac{b-a}{2^n}$

◦ Number of Iteration required to find desired solution
by BISECTION METHOD.

$$\frac{b-a}{2^n} < \epsilon$$

$$\Rightarrow \frac{2^n}{b-a} > \frac{1}{\epsilon}$$

$$\Rightarrow 2^n > \frac{b-a}{\epsilon}$$

$$\Rightarrow \boxed{n > \log_2 \left(\frac{b-a}{\epsilon} \right)}$$

Ques. 1) find real root of $f(x) = x^3 - 2x - 5$

Ans. 1) (Hit & Trial)

$$f(0) \rightarrow -ve.$$

$$f(2) \rightarrow -ve.$$

$$f(1) \rightarrow -ve.$$

$$f(3) \rightarrow +ve.$$

So,

$$a = 2$$

$$b = 3$$

$$\longrightarrow [a, b] = [2, 3]$$

• 1st Iteration

$$x_0 = \frac{a+b}{2} = \frac{2+3}{2}$$

$$\Rightarrow x_0 = 2.5$$

Now,

$$f(x_0) = 5.625 > 0$$

(+ve)

So,

Next approx. root will lie in interval $[2, 2.5]$

• 2nd Iteration

$$x_1 = \frac{a+x_0}{2} = \frac{2+2.5}{2}$$

$$\Rightarrow x_1 = 2.25$$

Now,

$$f(x_1) = 1.89062 > 0$$

(+ve)

So,

Next root in interval : $[2, 2.25]$

• 3rd Iteration

$$x_2 = \frac{a+x_1}{2} = \frac{2+2.25}{2}$$

$$\Rightarrow x_2 = 2.125$$

Now,

$$f(x_2) \rightarrow +ve$$

So,

Next root in interval : $[2, 2.125]$

• 4th Iteration

$$x_3 = \frac{2+2.125}{2} = 2.0625$$

$$\Rightarrow f(x_3) = -ve$$

So,

Next approx. root in the interval : $[2.0625, 2.125]$

• 5th Iteration

$$x_4 = \frac{2.0625+2.125}{2}$$

$$\Rightarrow x_4 = 2.09375$$

$$\Rightarrow f(x_4) \rightarrow -ve$$

So,

Next approx. root in interval : $[2.09375, 2.125]$

• 6th Iteration

$$x_5 = \frac{2.09375+2.125}{2}$$

$$\Rightarrow x_5 = 2.109375$$

Ques 2) $f(x) = x \sin x + \cos x$. Using Bisection method find root of $f(x)$.

Ans 2) $f(0) = 1$ (+ve) $f(2) \rightarrow 1.40244$ (+ve)
 $f(1) = 1.38177$ (+ve) $f(3) \rightarrow -0.5666$ (-ve)

So, $a = 2$ $\rightarrow$ interval : $[2, 3]$
 $b = 3$

• <u>1st Iteration</u> $x_0 = \frac{a+b}{2} = \frac{2+3}{2}$ $\Rightarrow x_0 = 2.5$ Now, $f(x_0) \rightarrow +ve$ So, Next Interval of approx. root : $(2.5, 3)$	• <u>2nd Iteration</u> $x_1 = \frac{2.5+3}{2}$ $\Rightarrow x_1 = 2.75$ Now, $f(x_1) \Rightarrow +ve$ So, Next approx. root in the interval : $(2.75, 3)$	• <u>3rd Iteration</u> $x_2 = \frac{2.75+3}{2}$ $\Rightarrow x_2 = 2.875$ Now, $f(x_2) \Rightarrow -ve$ So, Next approx. root in the interval : $(2.75, 2.875)$
• <u>4th Iteration</u> $x_3 = \frac{2.75 + 2.875}{2}$ $\Rightarrow x_3 = 2.8125$ Now, $f(x_3) \rightarrow -ve$ So, Next Interval of approx. root : $(2.75, 2.8125)$	• <u>5th Iteration</u> $x_4 = \frac{2.75 + 2.8125}{2}$ $\Rightarrow x_4 = 2.78125$ Now, $f(x_4) \rightarrow +ve$ So, Next approx. root in the interval : $(2.78125, 2.8125)$	• <u>6th Iteration</u> $x_5 = \frac{2.78125 + 2.8125}{2}$ $\Rightarrow x_5 = 2.796875$

CONVERGENCE ANALYSIS OF BISECTION METHOD

- $f(x) = 0$ (given function)
- $x = \alpha$ is actual root of $f(x)$.
- $x_n \rightarrow$ is n^{th} approx. root / successive approximation of actual root $x = \alpha$.

$\rightarrow x_n$ is said to converge to actual root $x = \alpha$ with order $q \geq 1$ if $|x_{n+1} - \alpha| \leq C|x_n - \alpha|^q$

$$|x_{n+1} - \alpha| \leq C|x_n - \alpha|^q$$

$C \rightarrow$ Rate of Convergence.
 $q \rightarrow$ Order of Convergence.

Note : When $q=1$ & $0 < C < 1$, then convergence is said to be of 1st Order.

- Order of Convergence $\rightarrow q$.
- Rate of Convergence $\rightarrow C$.

BISECTION METHOD ALWAYS CONVERGES.

$\rightarrow$ let ' α ' be actual root of $f(x) = 0 \Rightarrow f(\alpha) = 0$.

$\rightarrow$ let x_n is successive approximation of actual root α .

$$x_n = \epsilon_n + \alpha$$

$\epsilon \rightarrow$ Error (gap b/w actual & approx. root)

and, $x_{n-1} = \epsilon_{n-1} + \alpha$ ——— ①

So,

$$x_{n+1} = \frac{x_{n-1} + x_n}{2}$$

$$\Rightarrow x_{n+1} = \frac{\epsilon_{n-1} + \alpha + \epsilon_n + \alpha}{2}$$

$$\Rightarrow \epsilon_{n+1} + \alpha = \frac{\epsilon_{n-1} + \epsilon_n}{2} + \alpha$$

$$\Rightarrow \epsilon_{n+1} = \frac{\epsilon_n}{2} \left[1 + \frac{\epsilon_{n-1}}{\epsilon_n} \right]$$

$\left(\because \frac{\epsilon_{n-1}}{\epsilon_n} \text{ is v.v. small, we can neglect it.} \right)$

$$\Rightarrow \epsilon_{n+1} \approx \frac{\epsilon_n}{2}$$

$$\Rightarrow \boxed{\epsilon_{n+1} = \frac{1}{2} (\epsilon_n)^1}$$

So,

Rate of Convergence = $\frac{1}{2}$
Order of Convergence = 1

NEWTON - RAPHSOON METHOD

→ let x_0 be approximate root of f^n $f(x)$. And, $x_1 = x_0 + h$ be the correct root, then :-

$$\begin{aligned} f(x_1) &= 0 \\ \Rightarrow f(x_0 + h) &= 0 \end{aligned}$$

$$\Rightarrow f(x_0) + h f'(x_0) + \frac{h^2}{2!} f''(x_0) + \dots = 0$$

$$\Rightarrow f(x_0) + h f'(x_0) = 0 \quad (\text{Neglecting higher powers})$$

$$\Rightarrow h = - \frac{f(x_0)}{f'(x_0)}$$

So,

$$x_1 = x_0 + h$$

$$\Rightarrow x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

So,

In General ,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \quad (n \rightarrow 0, 1, 2, \dots)$$

Ques.1.) Use Newton - Raphson method to find approx. root of $(30)^{-1/5}$.

Ans.) $x = (30)^{-1/5}$
 $\Rightarrow x = \left(\frac{1}{30}\right)^{1/5}$

$$\Rightarrow x^5 = \frac{1}{30}$$

so,

$$f(x) = x^5 - \frac{1}{30}$$

$$\Rightarrow f(x) = 30x^5 - 1$$

so,

$$\text{let, } x_0 = \frac{1}{2} \quad \left(\text{Choose } x_0 \text{ such that } f(x) \text{ is nearly equal to zero} \right)$$

so,

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \quad (n \rightarrow 0, 1, 2, \dots)$$

$n=0$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$\Rightarrow x_1 = \frac{1}{2} - \frac{30x_0^5 - 1}{150x_0^4}$$

$$\Rightarrow x_1 = \frac{1}{2} - \frac{30\left(\frac{1}{2}\right)^5 - 1}{150\left(\frac{1}{2}\right)^4}$$

$$\Rightarrow x_1 = \frac{1}{2} + \frac{0.0625}{9.375}$$

$$\Rightarrow x_1 = 0.50667$$

$n=1$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$\Rightarrow x_2 = 0.50667 - \frac{30(0.50667)^5 - 1}{150(0.50667)^4}$$

$$\Rightarrow x_2 = 0.50667 - \frac{0.001689}{9.885}$$

$$\Rightarrow x_2 = 0.50649$$

$$n=2$$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

$$\Rightarrow x_3 = 0.50649 - \frac{30(0.50649)^5 - 1}{150(0.50649)^4}$$

$$\Rightarrow x_3 = 0.50649 + \frac{0.0005611}{9.87131}$$

$$\Rightarrow x_3 = 0.50654$$

Ques. 2) Use N-R method to find approximate root of $(24)^{1/3}$.

Ans.) let, $x = (24)^{1/3}$

$$\Rightarrow x^3 = 24$$

So,

$$f(x) = x^3 - 24$$

let, $x_0 = 3$

We know :- $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$

$$n=0$$

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)} = x_0 - \frac{x^3 - 24}{3x^2}$$

$$\Rightarrow x_1 = 3 - \frac{(3)^3 - 24}{3(3)^2}$$

$$\Rightarrow x_1 = 2.88889$$

$n=1$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$\Rightarrow x_2 = 2.8889 - \frac{(2.8889)^3 - 24}{3(2.8889)^2}$$

$$\Rightarrow x_2 = 2.8889 - \frac{0.10974}{25.037}$$

$$\Rightarrow x_2 = 2.8845$$

$n=2$

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

$$\Rightarrow x_3 = 2.8845 - \frac{(2.8845)^3 - 24}{3(2.8845)^2}$$

$$\Rightarrow x_3 = 2.884499$$

Ques: Use N-R method to derive formula for $N^{1/9}$, $N > 0$ & hence find value of $18^{1/3}$ correct to 4 decimal places.

Ans: let, $x = N^{1/9}$

$$\Rightarrow x^9 = N$$

$$\Rightarrow x^9 - N = 0$$

so,

$$f(x) = x^9 - N$$

$$\Rightarrow f(x_n) = x_n^9 - N$$

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$\Rightarrow x_{n+1} = x_n - \frac{x_n^3 - 18}{3x_n^2}$$

(General formula)

Now,

So,

$$x = 18^{1/3}$$

$$\Rightarrow x^3 = 18$$

$$\Rightarrow x^3 - 18 = 0$$

So,

$$f(x) = x^3 - 18$$

$$\Rightarrow f(x_n) = x_n^3 - 18$$

$$n=0$$

$$x_{n+1} = x_n - \frac{x_n^3 - 18}{3x_n^2}$$

$$x_1 = x_0 - \frac{x_0^3 - 18}{3x_0^2}$$

$$\text{let, } x_0 = 2.5$$

$$\Rightarrow x_1 = 2.5 - \frac{(2.5)^3 - 18}{3(2.5)^2}$$

$$\Rightarrow x_1 = 2.5 + 0.12667$$

$$\Rightarrow x_1 = 2.62667$$

$$n=1$$

$$x_2 = x_1 - \frac{x_1^3 - 18}{3x_1^2}$$

$$\Rightarrow x_2 = (2.62667) - \frac{(2.62667)^3 - 18}{3(2.62667)^2}$$

$$\Rightarrow x_2 = 2.62093$$

$n=2$

$$x_3 = x_2 - \frac{(x_2)^3 - 18}{3x_2^2}$$

$$\Rightarrow x_3 = 2.62093 - \left(\frac{(2.62093)^3 - 18}{3(2.62093)^2} \right)$$

$$\Rightarrow x_3 = 2.62074$$

$n=3$

$$x_4 = x_3 - \frac{x_3^3 - 18}{3x_3^2}$$

$$\Rightarrow x_4 = (2.62074) - \left(\frac{(2.62074)^3 - 18}{3(2.62074)^2} \right)$$

$$\Rightarrow x_4 = 2.62074$$

CONVERGENCE ANALYSIS OF NEWTON-RAPHSON METHOD

we know : $\epsilon_n = x_n - \alpha$ ① ($\alpha \rightarrow$ Actual root
 $f(x) = 0$)

So, $\epsilon_{n+1} = x_{n+1} - \alpha$

$$\Rightarrow \epsilon_{n+1} = \left(x_n - \frac{f(x_n)}{f'(x_n)} \right) - \alpha$$

$$\left(\begin{array}{l} x_n = \epsilon_n + \alpha \\ \text{(from eqn. ①)} \end{array} \right)$$

$$\Rightarrow \epsilon_{n+1} = \left(\epsilon_n + \alpha - \frac{f(\epsilon_n + \alpha)}{f'(\epsilon_n + \alpha)} \right) - \alpha$$

$$\Rightarrow E_{n+1} = E_n - \frac{f(x+E_n)}{f'(x+E_n)}$$

Using Taylor's expansion, we get,

$$f(x+h) = f(x) + h f'(x) + \frac{h^2}{2!} f''(x) + \dots$$

$$\Rightarrow E_{n+1} = E_n - \left(\frac{f(x) + E_n f'(x) + \frac{E_n^2}{2!} f''(x) + \dots}{f'(x) + E_n f''(x) + \frac{E_n^2}{2!} f'''(x) + \dots} \right)$$

(neglecting higher power as they are v.v. small)

$$\Rightarrow E_{n+1} = E_n - \left(\frac{f(x) + E_n f'(x) + \frac{E_n^2}{2!} f''(x)}{f'(x)} \right)$$

$$\Rightarrow E_{n+1} = E_n - \frac{f(x)}{f'(x)} - \frac{E_n f'(x)}{f'(x)} - \frac{E_n^2}{2!} \frac{f''(x)}{f'(x)}$$

$$\Rightarrow E_{n+1} = \cancel{E_n} - \cancel{E_n} - \frac{E_n^2}{2!} \frac{f''(x)}{f'(x)} \rightarrow (\text{constant} = K)$$

$$\Rightarrow E_{n+1} = K E_n^2$$

so, NR method has 2nd Order Convergence.

UNIT - 1

FIBONACCI SEARCH METHOD

($\forall \rightarrow$ for all)

FIBONACCI NUMBERS

$$F_0 = F_1 = 1$$

$$F_n = F_{n-1} + F_{n-2}$$

$$\forall n \geq 2$$

so,

$$\begin{matrix} 1 & , & 1 & , & 2 & , & 3 & , & 5 & , & 8 & , & 13 & , & 21 & , & 34 & , & \dots \\ F_0 & & F_1 & & F_2 & & F_3 & & F_4 & & F_5 & & F_6 & & F_7 & & F_8 \end{matrix}$$

→ In this method, we reduce search interval according to fibonacci Number.

Procedure

- (1) Choose, lower Bound = a
Upper Bound = b
and, set, $L = b - a$

Assume the desired no. of function evaluation to be n , set $k=2$.

- (2) Compute : $L_k^* = \left(\frac{F_{n-k+1}}{F_{n+1}} \right) \cdot L$

$$x_1 = a + L_k^* \quad \leftarrow \quad x_2 = b - L_k^*$$

3. Compute $f(x_1)$ and $f(x_2)$.

Use fundamental region - elimination rule to eliminate a region.

Set new 'a' & 'b'.

4. Check: Is $K=n$?

- If no, $\Rightarrow$ set $K=K+1$ & go to step 2.
- Else $\Rightarrow$ Terminate.

Ques. 1) Minimize the function $f(x) = x^2 + \frac{5}{x}$ in $(0, 5)$.

Ans.)

$$K=2$$

& taking $n=3$

1st Iteration

$$\textcircled{1} \quad \begin{array}{l} a=0 \\ b=5 \end{array} \Rightarrow \begin{array}{l} L=b-a \\ =5 \end{array}$$

so,

$$\textcircled{2} \quad L_k^* = \left(\frac{f_{n-k+1}}{f_{n+1}} \right) L$$

$$\Rightarrow L_k^* = \left(\frac{f_2}{f_4} \right) \times 5$$

$$\Rightarrow L_k^* = \frac{2}{5} \times 5$$

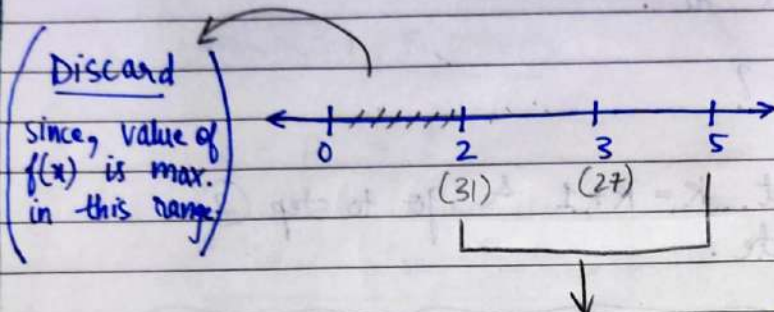
$$\Rightarrow L_k^* = 2$$

$$\textcircled{3} \quad \begin{array}{l} x_1 = a + L_k^* \\ = 0 + 2 \\ = 2 \end{array}$$

$$\begin{array}{l} x_2 = b - L_k^* \\ = 5 - 2 \\ = 3 \end{array}$$

③ $f(x_1) = 2^2 + \frac{54}{2} = 31$

& $f(x_2) = 3^2 + \frac{54}{3} = 27$



So, new interval : $(2, 5)$

New $a = 2$, $b = 5$

and, $k = k+1 = 3$ $\left(\because k \neq n \right)$
 $(2 \neq 3)$

Iteration 2nd

① $L = b - a = 5 - 2 = 3$

② $L_k^* = \left(\frac{f_{3-3+1}}{f_{3+1}} \right) \times L = \left(\frac{f_1}{f_4} \right) (3)$

$\Rightarrow L_k^* = \frac{1}{5} \times 3$

$\Rightarrow L_k^* = \frac{3}{5} = 0.6$

So,

$x_1 = a + L_k^* = 2 + 0.6$

Δ Notebook = 2.6

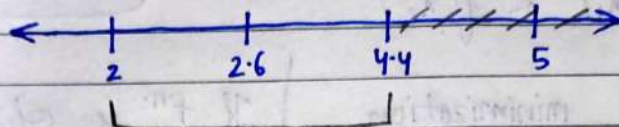
$x_2 = b - L_k^* = 5 - 0.6 = 4.4$

③ $f(x_1) = f(2.6)$
 $= 27.529$

$f(x_2) = f(4.4)$
 $= 31.6327$

so,

DISCARD.



So, New Interval : $(2, 4.4)$

$a = 2$

$b = 4.4$

but, since $n = k$, We shall
(3=3)

Stop here.

GOLDEN SECTION METHOD

Golden Ratio $\Rightarrow \left(\begin{array}{l} 1 : 1.618 = \phi \\ \text{OR} \\ \frac{1}{\phi} = 0.618 \end{array} \right)$

$\phi = 1.618 \Rightarrow \frac{1}{\phi} = 0.618$

Iteration Process

① Initial Interval of Uncertainty $[a, b]$

$b_0 = b - a$

② $L_2^* = \frac{1}{r^2} \cdot L_0$

then,

$$x_1 = a + L_2^*$$

$$x_2 = b - L_2^*$$

check, $f(x_1) < f(x_2)$

If f^n is of minimization
 $\Rightarrow$ Discard $[x_2, b]$

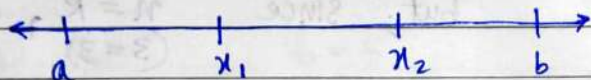
In General :

$$L_n^* = \frac{1}{r^n} \cdot L_0$$

$$L_n = \frac{1}{r^{n-1}} \cdot L_0$$

If f^n is of maximization
 $\Rightarrow$ Discard $[a, x_1]$

③



$$L_3^* = \frac{1}{r^3} \cdot L_0$$

and then,

$$x_3 = b - L_3^*$$

(If you discarded $[a, x_1]$)

$$x_3 = a + L_3^*$$

(If you discarded $[x_2, b]$)

④ Check for STOPPING TOLERANCE.

$$\epsilon = 0.15 \quad (\text{Stopping Tolerance})$$

$\rightarrow$ If, value of $L_n = \left(\frac{1}{r^{n-1}}\right) \cdot L_0$ in a particular iteration comes equal to $\epsilon = 0.15 \therefore$ TERMINATE.

Ques. 1.) Minimise $f(x) = 4x^3 + x^2 - 7x + 14$ within interval $(0, 1)$ using Golden Section method.

Ans.) • 1st Iteration

$$L_0 = b - a = 1 - 0 = 1$$

• 2nd Iteration

$$L_2^* = \frac{1}{\phi^2} \cdot L_0 = \frac{1}{(1.618)^2} \cdot 1 = 0.3819$$

so,

$$x_1 = a + L_2^* = 0.3819$$

$$x_2 = b - L_2^* = 0.618$$

so,

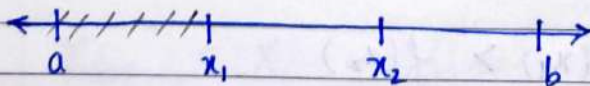
$$f(x_1) = f(0.3819) = 11.6953$$

$$f(x_2) = f(0.618) = 11.000$$

so,

$$f(x_1) > f(x_2)$$

so, discard $\therefore [a, x_1]$



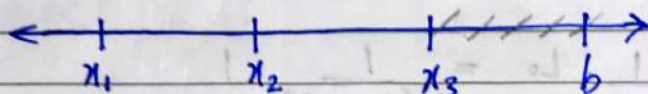
• 3rd Iteration

$$L_3^* = \frac{1}{\phi^3} \cdot L_0 = \frac{1}{(1.618)^3} \cdot 1 = 0.236$$

So, $x_3 = b - L_3^* = 1 - 0.236$
 $= 0.764$

So, $f(x_3) = f(0.764)$
 $= 11.0193$

$\therefore f(x_3) > f(x_2)$
 So, discard: $[x_3, b]$



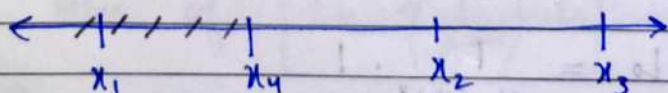
• 4th Iteration

$L_4^* = \frac{1}{x^4} \cdot 10 = \frac{1}{(1.618)^4} \cdot 1$
 $= 0.1459$

So, $x_4 = x_1 + L_4^* = 0.3819 + 0.1459$
 $= 0.5278$

So, $f(x_4) = f(0.5278)$
 $= 11.1721$

$\therefore f(x_4) > f(x_2)$
 discard: $[x_1, x_4]$



• 5th Iteration

→ Before proceeding with normal procedure, let's first check for Stopping Tolerance.

So,

$$L_5 = \frac{1}{2^4} \cdot L_0 = 0.1459$$

$$= 0.1459 \approx (\epsilon = 0.15)$$

So, we shall stop here.

NEWTON'S METHOD

$$1. \quad X_{i+1} = X_i - [J_i]^{-1} \cdot \nabla f_i$$

$J_i \rightarrow \text{Hessian Matrix}$
 $J \rightarrow \nabla^2 f$
 $\nabla f \rightarrow \text{Gradient of } f$

2. The Iterative formula for finding roots of eqⁿ.
 $f(x) = 0$ is

$$X_{i+1} = X_i - \frac{f(X_i)}{f'(X_i)}$$

(Iterative formula to find root of $f(x) = 0$)

3. This method always converges & has quadratic Rate of convergence.

④ Stopping Criteria : Stop when $\nabla f_i = \text{Initial Value}$
 x_1

Ques. 1.) Minimize $f(x_1, x_2) = x_1 - x_2 + 2x_1^2 + 2x_1x_2 + x_2^2$
Starting from point $x_1 = (0, 0)$.

Ans.: The gradient of f is ∇f :

means ∇f_{x_1}

$$\nabla f_1 = \begin{bmatrix} \partial f / \partial x_1 \\ \partial f / \partial x_2 \end{bmatrix}_{x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}} = \begin{bmatrix} 1 + 4x_1 + 2x_2 \\ -1 + 2x_1 + 2x_2 \end{bmatrix}_{x_1} = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Now,

Hessian Matrix, $J = \begin{bmatrix} \partial^2 f / \partial x_1^2 & \partial^2 f / \partial x_1 \partial x_2 \\ \partial^2 f / \partial x_2 \partial x_1 & \partial^2 f / \partial x_2^2 \end{bmatrix}$

so, $J_1 = \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix}$ (J_1 means J (at x_1) or J_{x_1})

so, $J_1^{-1} = \frac{1}{4} \begin{bmatrix} 2 & -2 \\ -2 & 4 \end{bmatrix} = \begin{bmatrix} 1/2 & -1/2 \\ -1/2 & 1 \end{bmatrix}$

• 1st Iteration

(put, $i = 1$)

so, $x_2 = x_1 - [J_1]^{-1} \nabla f_1$

$$\Rightarrow x_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 1/2 & -1/2 \\ -1/2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

$$\Rightarrow x_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} - \begin{bmatrix} 1 \\ -1.5 \end{bmatrix}$$

$$\Rightarrow x_2 = \begin{bmatrix} -1 \\ 1.5 \end{bmatrix}$$

Now,

Checking whether $\nabla f_2 = 0$ or not (Stopping Criteria)

$$\nabla f_2 = \nabla f_{x_2}$$

$$\Rightarrow \nabla f_2 = \begin{bmatrix} 1 + 4x_1 + 2x_2 \\ -1 + 2x_1 + 2x_2 \end{bmatrix}_{x_2 = (-1, 1.5)}$$

$$\Rightarrow \nabla f_2 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow \nabla f_2 = x_1 \quad (\text{So, we shall STOP.})$$

STEEPEST DESCENT METHOD

- ① Start with arbitrary initial point x_1 . Set Iteration.
- ② Find the search direction s_i

$$s_i = -\nabla f_i$$

(for $i=1$)

$$s_1 = -(\nabla f_{x_1})$$

$$\left(\text{where, } \nabla f_{x_1} = \begin{bmatrix} \partial f / \partial x_1 \\ \partial f / \partial x_2 \end{bmatrix}_{x_1 = (x_1, x_2)} \right)$$

③ Find the Optimal Step length λ_1^* .

$$\text{Minimize } f(x_1 + \lambda_1^* s_1) \Rightarrow \frac{df}{d\lambda_1} = 0 \Rightarrow \lambda_1^* = ?$$

④ $x_2 = x_1 + \lambda_1^* s_1$

find ∇f_2 . If $\nabla f_2 = \begin{bmatrix} \partial f / \partial x_1 \\ \partial f / \partial x_2 \end{bmatrix}_{x_2} = \text{Initial Point } (x_1)$

then, STOP.

Ques. 1.) Minimize $f(x_1, x_2) = x_1 - x_2 + 2x_1^2 + 2x_1x_2 + x_2^2$
Starting from pt. $x_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$.

Ans. 1.) • 1st Iteration

$$\nabla f_1 = \begin{bmatrix} \partial f / \partial x_1 \\ \partial f / \partial x_2 \end{bmatrix} = \begin{bmatrix} 1 + 4x_1 + 2x_2 \\ -1 + 2x_1 + 2x_2 \end{bmatrix}_{x_1 = (0, 0)}$$

$$\Rightarrow \nabla f_1 = \begin{bmatrix} 1 + 4(0) + 2(0) \\ -1 + 2(0) + 2(0) \end{bmatrix}$$

$$\Rightarrow \nabla f_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

so,

$$s_1 = -\nabla f_1$$

$$\Rightarrow s_1 = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

Now,

$$X_2 = X_1 + \lambda_1 S_1 = \begin{bmatrix} 0 \\ 0 \end{bmatrix} + \lambda_1 \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 - \lambda_1 \\ 0 + \lambda_1 \end{bmatrix} = \begin{bmatrix} -\lambda_1 \\ \lambda_1 \end{bmatrix} \rightarrow \begin{matrix} x_1 \\ x_2 \end{matrix}$$

so,

$$\begin{aligned} f(x_1 + \lambda_1 S_1) &= -\lambda_1 - \lambda_1 + 2\lambda_1^2 - 2\lambda_1^2 + \lambda_1^2 \\ &= -2\lambda_1 + \lambda_1^2 \end{aligned}$$

so,

$$\frac{df}{d\lambda_1} = 0$$

$$\Rightarrow 2\lambda - 2 = 0$$

$$\Rightarrow \lambda = 1 \xrightarrow{\text{so}} X_2 = \begin{bmatrix} -\lambda_1 \\ \lambda_1 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

• 2nd Iteration

$$\nabla f_2 = \begin{bmatrix} \partial f / \partial x_1 \\ \partial f / \partial x_2 \end{bmatrix} = \begin{bmatrix} 1 + 4x_1 + 2x_2 \\ -1 + 2x_1 + 2x_2 \end{bmatrix} \quad x_2 = \begin{bmatrix} -1 \\ -1 \end{bmatrix}$$

$$\Rightarrow \nabla f_2 = \begin{bmatrix} 1 - 4 + 2 \\ -1 - 2 + 2 \end{bmatrix}$$

$$\Rightarrow \nabla f_2 = \begin{bmatrix} -1 \\ -1 \end{bmatrix} \neq 0 \quad \therefore \text{So, Continue.}$$

so,

$$S_2 = -\nabla f_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Now, $X_3 = X_2 + \lambda_2 S_2$

$$\Rightarrow X_3 = \begin{bmatrix} -1 \\ 1 \end{bmatrix} + \lambda_2 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\Rightarrow X_3 = \begin{bmatrix} -1 + \lambda_2 \\ 1 + \lambda_2 \end{bmatrix}$$

sq $f(X_2 + \lambda_2 S_2) = (-1 + \lambda_2) - (1 + \lambda_2) + 2(-1 + \lambda_2)^2 + 2(-1 + \lambda_2)(1 + \lambda_2) + (1 + \lambda_2)^2$

$$\Rightarrow f(X_2 + \lambda_2 S_2) = -1 + \lambda_2 - 1 - \lambda_2 + 2 - 4\lambda_2 + 2\lambda_2^2 + 2\lambda_2^2 - 2 + 1 + 2\lambda_2 + \lambda_2^2$$

$$\Rightarrow f(X_2 + \lambda_2 S_2) = 5\lambda_2^2 - 2\lambda_2 - 1$$

sq $\frac{df}{d\lambda_2} = 0$

$$\Rightarrow 10\lambda_2 - 2 = 0$$

$$\Rightarrow \lambda_2 = \frac{2}{10} = \frac{1}{5}$$

$$\Rightarrow \lambda_2 = 0.2$$

sq $\Rightarrow X_3 = \begin{bmatrix} -1 + 0.2 \\ 1 + 0.2 \end{bmatrix} = \begin{bmatrix} -0.8 \\ 1.2 \end{bmatrix}$

$$\therefore \begin{bmatrix} -0.8 \\ 1.2 \end{bmatrix} = X_3$$

• 3rd Iteration

$$\nabla f_3 = \begin{bmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{bmatrix}_{X_3} = \begin{bmatrix} 1 + 4x_1 + 2x_2 \\ -1 + 2x_1 + 2x_2 \end{bmatrix}_{X_3} = \begin{bmatrix} 1 + 4(-0.8) + 2(1.2) \\ -1 + 2(-0.8) + 2(1.2) \end{bmatrix}$$

$$\rightarrow \nabla f_3 = \begin{bmatrix} 0.2 \\ -0.2 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix} \text{ } x_1 (\text{initial pt.})$$

So,

$$x_4 = x_3 + \lambda_3 S_3$$

$$= \begin{bmatrix} -0.8 \\ 1.2 \end{bmatrix} + \lambda_3 \begin{bmatrix} -0.2 \\ 0.2 \end{bmatrix}$$

$$= \begin{bmatrix} -0.8 - \lambda_3(0.2) \\ 1.2 + \lambda_3(0.2) \end{bmatrix}$$

$$(S_3 = -\nabla f_3 = \begin{bmatrix} -0.2 \\ 0.2 \end{bmatrix})$$

So,

$$f(x_3 + \lambda_3 S_3) = 0.04 \lambda_3^2 - 0.08 \lambda_3 - 1.2$$

So,

$$\frac{df}{d\lambda_3} = 0$$

$$\rightarrow 0.08 \lambda_3 - 0.08 = 0$$

$$\rightarrow \lambda_3 = 1 \xrightarrow{\text{So,}}$$

$$x_4 = \begin{bmatrix} -0.8 - 0.2(1) \\ 1.2 + 0.2(1) \end{bmatrix}$$

$$\Rightarrow x_4 = \begin{bmatrix} -1 \\ 1.4 \end{bmatrix}$$

Now,

$$\nabla f_4 = \begin{bmatrix} \partial f / \partial x_1 \\ \partial f / \partial x_2 \end{bmatrix}_{x_4} = \begin{bmatrix} 0.2 \\ 0.2 \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Continue.

• 4th Iteration

$$S_4 = -\nabla f_4 \\ = \begin{bmatrix} -0.2 \\ -0.2 \end{bmatrix}$$

$$\text{So, } X_5 = X_4 + \lambda_4 S_4 = \begin{bmatrix} -1 \\ 1.4 \end{bmatrix} + \lambda_4 \begin{bmatrix} -0.2 \\ -0.2 \end{bmatrix}$$

$$X_5 = \begin{bmatrix} -1 - 0.2\lambda_4 \\ 1.4 - 0.2\lambda_4 \end{bmatrix}$$

$$\text{So, } f(X_4 + \lambda_4 S_4) = (-1 - 0.2\lambda_4)^2 - (1.4 - 0.2\lambda_4)^2 + 2(-1 - 0.2\lambda_4)^2 \\ + 2(-1 - 0.2\lambda_4)(1.4 - 0.2\lambda_4) + (1.4 - 0.2\lambda_4)^2$$

$$= -1 - 0.2\lambda_4 - 1.4 + 0.2\lambda_4 + 0.08\lambda_4^2 + 0.8\lambda_4 + 2 \\ - 2.8 + 0.4\lambda_4 - 0.56\lambda_4 + 0.08\lambda_4^2 + 1.96 \\ - 0.56\lambda_4 + 0.04\lambda_4^2$$

$$= 0.2\lambda_4^2 + 0.08\lambda_4 - 1.24$$

$$\text{So, } \frac{df}{d\lambda_4} = 0$$

$$\Rightarrow 0.4\lambda_4 + 0.08 = 0$$

$$\Rightarrow \lambda_4 = \frac{-0.08}{0.4} = -0.2$$

$$\text{So, } X_5 = \begin{bmatrix} -1 - 0.2(-0.2) \\ 1.4 - 0.2(-0.2) \end{bmatrix} \\ = \begin{bmatrix} -0.96 \\ 1.44 \end{bmatrix}$$

$$\text{So, } \nabla f_5 = \begin{bmatrix} 1 + 4x_1 + 2x_2 \\ -1 + 2x_1 + 2x_2 \end{bmatrix} = \begin{bmatrix} 1 + 4(-0.96) + 2(1.44) \\ -1 + 2(-0.96) + 2(1.44) \end{bmatrix}$$

$$= \begin{bmatrix} 0.04 \\ -0.04 \end{bmatrix} \approx \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

(nearly equal to)

so, we shall stop.